# Project Plan

## Study Buddy – Smart Study Material Generator

**Team 25**

| Role | Name |
|---|---|
| Industry Partner | N/A |
| Primary Instructor | Prof. Anjana Shah |
| Team Lead / Full Stack Developer | Parsa Molahosseini |
| Backend & Integration Developer | Kevin George Buhain |
| QA & Documentation Lead | Mehrad Bayat |
| UI / UX Designer | Jerry-Lee Somera |

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### Document Revision History

| Revision # | Date | Changes |
|---|---|---|
| 1.0 | October 6, 2025 | Initial project plan release |
| 1.1 | February 6, 2026 | Milestones date changes and |

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### 1. Executive Summary

Study Buddy is a web-based platform that lets students upload their notes (PDF/TXT) and automatically generate flashcards, quizzes, and summaries. It improves study efficiency through automation, progress tracking, and spaced-review reminders. The platform aims to generate study materials 80% faster than manual creation, helping students optimize their learning workflow.

| Item | Details |
|---|---|
| Objective | Develop an automated study material generator to help students create flashcards, quizzes, and summaries from uploaded notes |
| **Corporate Goals** Addressed | Enhance student learning efficiency; Provide accessible educational technology tools; Reduce manual study material preparation time |
| Planned Start Date | September 2025 |
| Planned End Date | April 2026 |

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### 2. Project Approvers, Reviews and Distribution List

| Project Role | Name | E-mail |
|---|---|---|
| Supervisor | Prof. Anjana Shah | ashah@georgebrown.ca |
| Team Lead / Full Stack Developer | Parsa Molahosseini | parsa.molahosseini@georgebrown.ca |
| Backend & Integration Developer | Kevin George Buhain | kevingeorge.buhain@georgebrown.ca |
| QA & Documentation Lead | Mehrad Bayat | mehrad.bayat@georgebrown.ca |
| UI/UX Designer | Jerry-Lee Somera | jerry-lee.somera@georgenrown.ca |

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## 3. Scope

| In Scope | Out of Scope |
|---|---|
| File upload & text extraction (PDF/TXT) | AI integration (reserved for Phase 2) |
| Flashcard generation with customization | Native mobile applications (iOS/Android) |
| Quiz generation with multiple choice format | LMS integration (Canvas, Moodle, Blackboard) |
| Progress dashboard and analytics | Video content processing |
| Smart highlight extraction from notes | Handwriting recognition |
| Spaced-review reminders via email/notifications | Real-time collaboration features |

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## 4. Deliverables

| Deliverable | Description |
|---|---|
| Functional Web Application | Working prototype with all core features implemented (file upload, flashcard generation, quiz creation, dashboard) |
| Source Code Repository | Complete codebase with version control history on GitHub, including README and setup instructions |
| Documentation Package | Project summary, vision document, high-level requirements (HLR), user manual, and technical documentation |
| Final Presentation | Demo slide deck showcasing features, architecture, and project outcomes |

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## 5. Assumptions

This project makes the following assumptions:

- Students have access to digital notes in PDF or TXT format
- Users have reliable Internet access and use modern web browsers
- Target users are comfortable with web-based applications and basic file uploads
- Study materials provided by users are primarily in English language
- Users are willing to create accounts for progress tracking features
- The hosting platform (Heroku/Render) will provide sufficient free-tier resources for prototype development

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## 6. Dependencies

The following are the internal and external dependencies that will have to be acknowledged and addressed:

**Technical Dependencies:**

- PyPDF2 or Apache Tika for PDF text extraction functionality
- Chart.js library for data visualization in progress dashboard
- FastAPI framework for backend REST API development
- PostgreSQL database for persistent data storage
- Heroku or Render cloud platform for application deployment
- GitHub for version control and team collaboration

**Team Dependencies:**

- Backend API completion required before frontend integration can begin
- UI mockups and design approval needed before implementation phase
- Regular team availability for weekly sprint meetings and code reviews
- Access to university network/resources for testing and development

**External Dependencies:**

- Availability of free cloud hosting services
- Stable internet connectivity for collaborative development
- Open-source library maintenance and compatibility

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## 7. Risk Management

| Potential Risk | Severity (H/M/L) | Likelihood (H/M/L) | Management Strategy |
|---|---|---|---|
| Parsing errors with complex PDF formats | M | L | Validate file types on upload; implement robust error handling; test with diverse file formats early; provide clear error messages to users |
| Time overrun on core features | H | M | Follow strict sprint planning with 2-week iterations; prioritize MVP features first; maintain 1-week buffer time in schedule; weekly progress reviews |
| Data loss or corruption | M | L | Implement regular automated backups (daily); use version control for all code; maintain database redundancy; implement proper transaction handling |
| Team member availability issues | M | M | Cross-train team members on multiple components; maintain comprehensive documentation; redistribute tasks as needed; hold regular knowledge-sharing sessions |
| Browser compatibility issues | M | L | Test on major browsers (Chrome, Firefox, Safari, Edge); use standard web APIs and CSS; implement graceful degradation for older browsers |
| Database performance with large files | M | M | Implement file size limits (max 10MB initially); optimize database queries with indexing; use pagination for large datasets; implement caching strategies |

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## 8. Communication

### Reporting

The following reports will be produced:

| Report | Audience | Frequency |
|---|---|---|
| Sprint Progress Report | Team & Instructor | Weekly (Mondays) |
| Milestone Status Update | Instructor | Bi-weekly (end of sprint) |
| Technical Documentation | Team & Stakeholder | As completed per milestone |
| Risk & Issue Log | Team & Instructor | Weekly (updated during team meetings) |

**Meetings**

The following meetings/communications will be established:

| Meeting | Purpose | Attendees | Frequency |
|---|---|---|---|
| Team Sprint Meeting | Review progress, plan upcoming tasks, resolve blockers, assign new work | All team members | Weekly (Mondays, 6:00 PM) |
| Professor Check-in | Project status update, milestone review, receive guidance and feedback | Team lead + Instructor | Bi-weekly (Fridays) |
| Code Review Sessions | Quality assurance, knowledge sharing, ensure coding standards | Development team | As needed (minimum bi-weekly) |
| Sprint Retrospective | Reflect on process, discuss improvements, celebrate wins | All team members | End of each 2-week sprint |

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## 9. Task Listing (WBS – Work Breakdown Structure)

| Reference | Tasks | Duration | Dependency |
|---|---|---|---|
| A | Requirements gathering & documentation (vision doc, HLR, user stories) | 2 weeks | - |
| B | UI design & mockups creation (wireframes, user flow diagrams) | 2 weeks | A |
| C | Backend setup (FastAPI framework, PostgreSQL database, API endpoints) | 3 weeks | B |
| D | Text extraction module development (PDF/TXT parsing, error handling) | 3 weeks | C |
| E | Flashcard & quiz generation logic (content parsing, question creation) | 3 weeks | D |
| F | Dashboard & highlights implementation (analytics, progress tracking) | 3 weeks | E |
| G | Testing, debugging & final documentation (user manual, bug fixes) | 2 weeks | F |

**Total Project Duration:** 18 weeks ( approximately 4.5 months )

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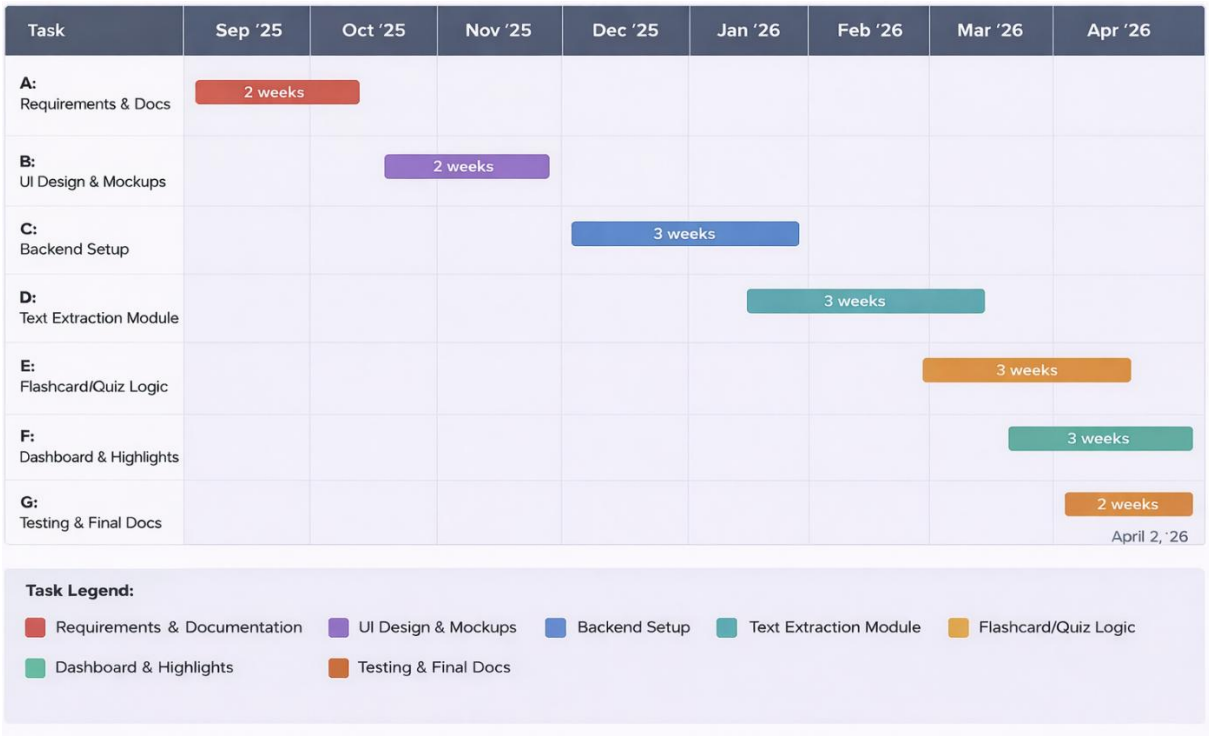
## 10. Gantt Chart




The Gantt chart above visualizes the project timeline from September 2025 through April 2026, showing task dependencies and parallel work streams.

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## 11. Milestones

| Major Activity or Milestone | Estimated Milestone Target Date | Owner/Reviewer Team Members |
|---|---|---|
| Project Proposal Approved | September 25, 2025 | Team 25 |
| UI Design & Backend Setup Ready | November 15, 2025 | Parsa, Kevin |
| Text Extraction Module Complete | December 15, 2025 | Parsa, Kevin |
| MVP (Minimum Viable Product) Completed | January 25, 2026 | Team 25 |
| Final Testing & QA Completed | February 20, 2026 | Mehrad |
| Final Demo & Presentation | April 2, 2026 | Jerry-Lee |

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## 12. RAM - Responsibility Assignment Matrix

| Task | Parsa | Kevin | Mehrad | Jerry-Lee |
|---|---|---|---|---|
| Architecture & Backend Development | A | R | C | I |
| Integration & System Testing | R | A | C | I |
| UI Design & Frontend Development | I | C | R | A |
| QA & Documentation | C | I | A | R |
| Project Planning & Coordination | A | C | I | R |
| Database Design & Implementation | A | R | I | C |
| Text Extraction Module | R | A | C | I |
| Deployment & DevOps | C | R | A | I |

Legend:

- **A** = Accountable (ultimate ownership and final approval)
- **R** = Responsible (does the work to complete the task)
- **C** = Consulted (provides input and expertise)
- **I** = Informed (kept updated on progress)

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## 13. Approval

The signatures below indicate their approval of the contents in this document.

| Project Role | Name | Signature | Date |
|---|---|---|---|
| Supervisor | Prof. Anjana Shah | | |
| Team Lead / Full Stack Developer | Parsa Molahosseini | | |
| Backend & Integration Developer | Kevin George Buhain | | |
| QA & Documentation Lead | Mehrad Bayat | | |
| UI/UX Designer | Jerry-Lee Somera | | |

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End of Document

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## Appendix A: Success Criteria

The project will be considered successful when:

- Users can successfully upload PDF/TXT files and receive generated flashcards
- Quiz generation functionality produces valid multiple-choice questions
- Dashboard displays accurate progress metrics and analytics
- System maintains 95%+ uptime during testing phase
- All core features pass user acceptance testing
- Documentation is complete and accessible

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## Appendix B: Technology Stack Summary

| Component | Technology |
|---|---|
| Backend Framework | FastAPI (Python) |
| Database | PostgreSQL |
| Frontend | HTML5, CSS3, JavaScript |
| Data Visualization | Chart.js |
| Text Extraction | PyPDF2 / Apache Tika |
| Hosting | Heroku or Render |
| Version Control | GitHub |

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**Document prepared by Team 25**

**Last updated: February 8, 2026**